

ATHENS, GA, USA

WMO: 723110

Lat: 33.948N

Lon: 83.328W

Elev: 239

StdP: 98.48

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13873

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.3	-3.2	-15.0	1.0	-0.1	-12.8	1.3	0.5	9.3	5.4	8.5	5.4	3.0	280	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	35.2	23.7	34.0	23.5	32.7	23.3	25.6	31.6	25.1	30.9	24.5	30.0	2.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.4	27.9	23.5	18.8	27.3	23.0	18.2	26.7	79.6	31.6	77.2	30.9	75.1	30.1	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	7.0	5.8	-9.2	37.0	2.4	2.0	-11.0	38.5	-12.4	39.7	-13.8	40.8	-15.6	42.3		
(5)			-10.4	26.7	2.2	1.1	-12.0	27.5	-13.2	28.2	-14.4	28.8	-16.0	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	6.6	8.7	12.6	16.8	21.2	25.2	26.7	26.3	23.1	17.4	11.7	8.1		
	DBStd	8.11	5.26	4.87	4.82	3.93	3.36	2.48	1.94	2.09	2.98	4.04	4.34	4.75		
	HDD10.0	371	133	76	31	3	0	0	0	0	0	2	31	96		
	HDD18.3	1499	364	271	186	73	14	1	0	0	4	66	202	318		
	CDD10.0	2953	28	39	110	207	348	454	519	504	394	231	82	36		
	CDD18.3	1038	0	1	7	28	104	205	261	246	147	36	3	1		
	CDH23.3	9382	1	5	69	301	932	1907	2507	2234	1137	268	22	2		
	CDH26.7	3941	0	0	8	59	307	859	1187	1024	437	57	1	0		
(14) Wind	WSAvg	2.5	3.0	3.0	3.1	2.7	2.3	2.1	2.0	1.9	2.2	2.3	2.4	2.7		
(15) Precipitation	PrecAvg	1238	117	113	134	95	103	98	122	88	92	79	90	98		
	PrecMax	1814	240	235	278	277	247	310	268	161	262	197	210	214		
	PrecMin	822	16	19	27	19	10	23	39	4	16	0	16	21		
	PrecStd	219	49	53	66	60	59	60	63	44	57	57	44	53		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	23.8	27.9	30.1	33.3	36.1	36.6	37.2	34.4	30.8	26.0	22.9		
		MCWB	15.3	16.4	17.3	19.5	21.3	23.2	23.9	23.9	22.4	21.5	18.2	16.6		
	2%	DB	19.3	21.1	25.3	28.4	31.5	34.4	35.1	35.1	32.7	28.0	23.4	20.4		
		MCWB	14.5	14.3	16.3	18.6	21.1	23.1	24.0	23.8	22.2	20.0	16.0	16.4		
	5%	DB	17.3	18.9	23.2	26.9	29.9	33.0	33.8	33.4	31.0	26.4	21.4	18.1		
		MCWB	13.8	13.4	15.0	17.8	20.9	23.0	24.0	23.7	22.1	18.8	15.6	14.8		
	10%	DB	15.2	16.9	21.0	25.0	28.3	31.4	32.4	31.8	29.4	24.4	19.3	16.1		
		MCWB	11.8	12.5	14.0	16.6	20.2	22.6	23.8	23.5	21.6	18.1	14.6	12.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.8	18.5	19.2	21.6	24.0	25.7	26.4	26.4	24.7	23.5	20.7	19.5		
		MCDB	19.9	21.1	24.1	27.1	28.8	31.8	32.8	32.5	30.0	26.6	22.3	20.5		
	2%	WB	16.3	16.9	17.8	20.3	23.0	24.9	25.7	25.6	24.0	22.0	18.8	17.5		
		MCDB	18.4	19.5	22.7	25.6	28.3	31.1	31.8	31.4	29.1	25.9	21.0	19.3		
	5%	WB	14.6	15.2	16.7	19.3	22.2	24.2	25.1	25.0	23.4	20.8	17.4	15.7		
		MCDB	16.6	17.6	20.9	24.2	27.4	29.9	31.0	30.5	28.0	23.8	19.8	17.8		
	10%	WB	12.4	13.4	15.5	18.2	21.4	23.6	24.6	24.4	22.7	19.8	15.7	13.4		
		MCDB	14.3	15.8	19.2	23.0	26.2	29.1	30.2	29.5	26.9	23.0	18.5	15.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	11.2	12.3	12.7	12.1	11.3	10.9	10.7	10.9	11.9	11.9	10.4		
		MCDBR	12.5	13.4	14.8	14.7	13.8	13.5	12.8	13.0	13.1	13.7	14.0	12.2		
	5% WB	MCWBR	8.4	8.2	7.4	6.4	5.2	4.3	3.9	4.1	4.6	6.3	7.7	8.1		
		MCWBR	10.2	10.6	12.0	12.0	11.3	11.1	11.0	10.8	10.4	10.2	10.3	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.313	0.319	0.355	0.390	0.451	0.480	0.520	0.518	0.422	0.383	0.341	0.310		
	taud		2.519	2.495	2.422	2.312	2.180	2.146	2.054	2.046	2.298	2.388	2.455	2.551		
	Ebn at Noon		902	933	920	898	842	814	779	773	840	848	857	885		
	Edh at Noon		83	94	110	129	149	153	167	166	123	104	88	77		
(44) All-Sky Solar Radiation	RadAvg		2.73	3.32	4.40	5.49	6.07	6.28	5.96	5.48	4.75	4.00	3.02	2.33		
	RadStd		0.16	0.40	0.31	0.37	0.48	0.48	0.41	0.28	0.46	0.47	0.29	0.24		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+155	+103
(47) Regional (42 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+112

Nomenclature: See separate page